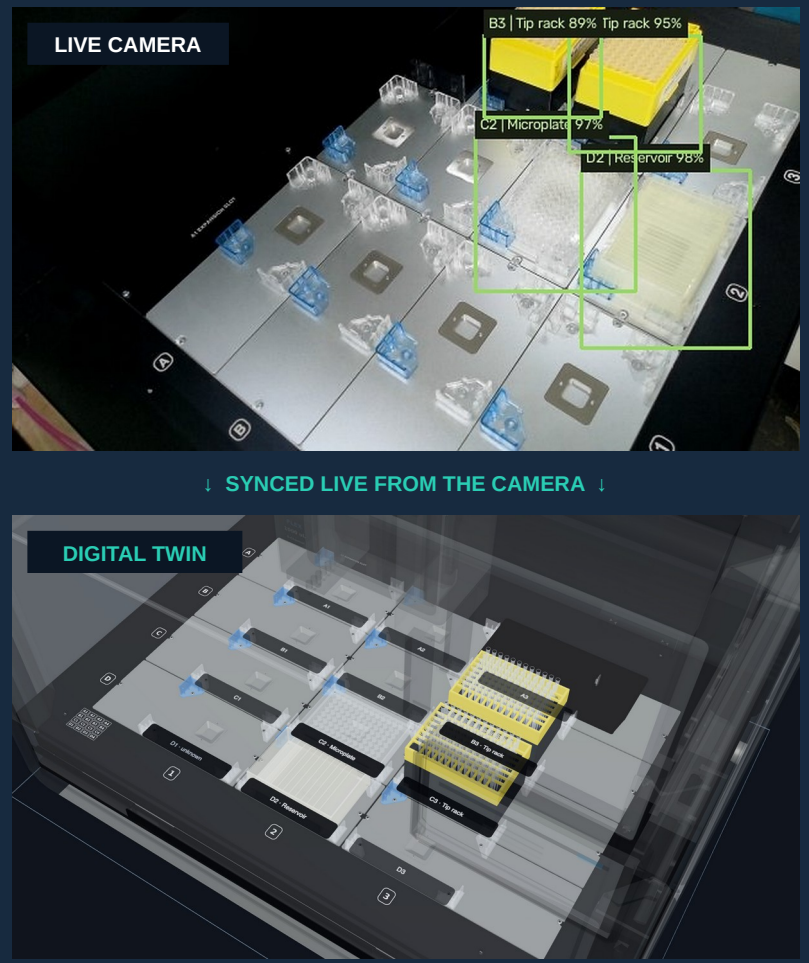
LAB AUTOMATION / VISUAL VERIFICATION

Lab Sentry

A live digital twin of the Opentrons deck that blocks run start until reality matches the protocol

3D twin of the Flex and its camera **generates the training data**, **mirrors the deck live**, and **gates the run**

PHYSICAL DECK ↔ DIGITAL TWIN



TEAM

Simon Coelho · YM Liaw · Saee Pangarkar
Barmak Modrek · Daniel Betz

github.com/simoncoelho/lab-sentry

01 / THE PROBLEM

Setup errors can go undetected until a run fails

Lab Sentry checks the deck before the protocol starts.

COMMON SETUP ERRORS

- Wrong or missing labware
- Labware in the wrong slot
- Extra labware left on the deck

Detects today: tip racks, microplates, reservoirs.
Next: lids, seals, used or empty tips.

THE SOLUTION

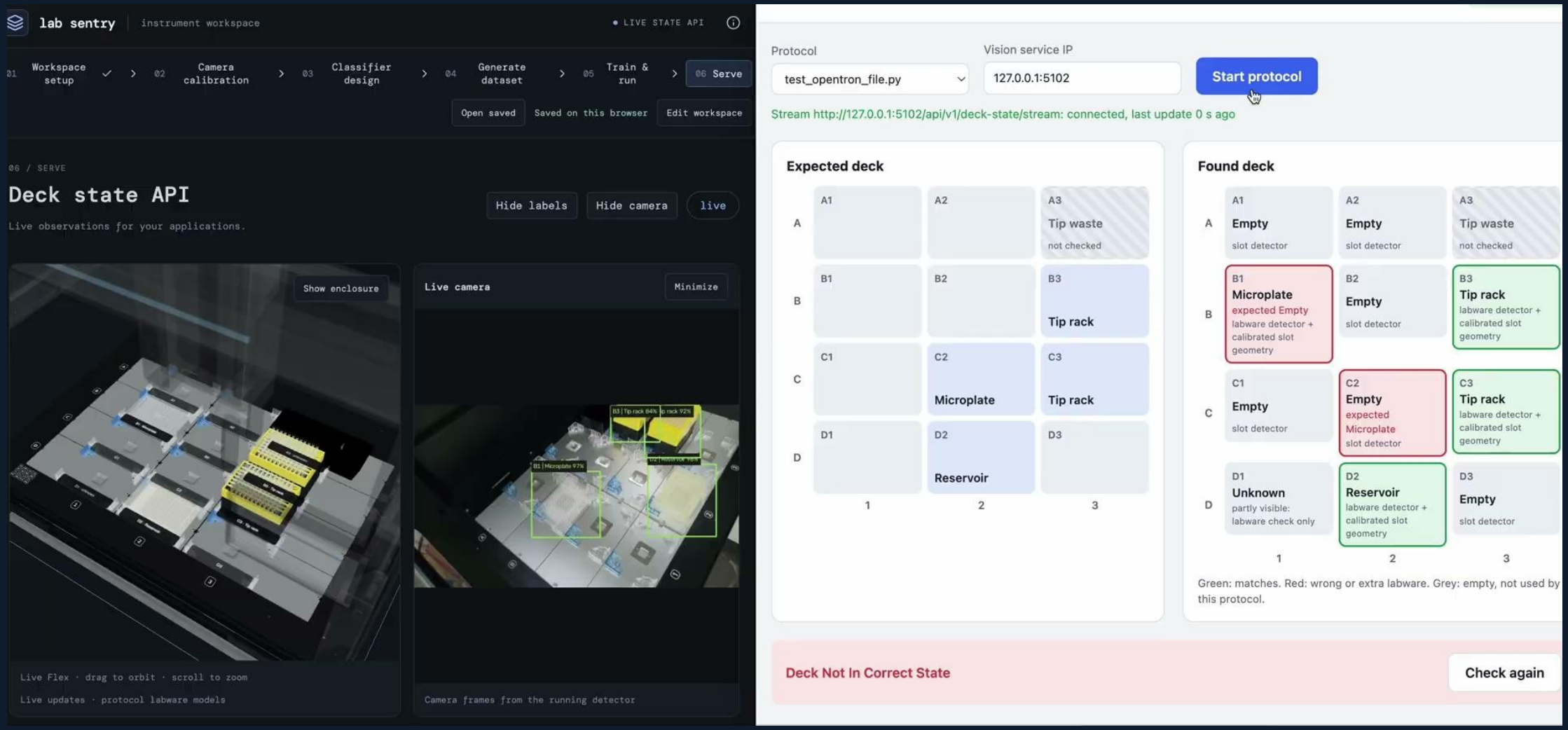
Verify the physical deck against the protocol before starting.

Detect mismatches early. Prevent avoidable failed runs.

Camera → Digital twin → Deck check → Run

INITIAL SCOPE • Pre-run verification; in-run monitoring is a future extension.

Protocol-aware deck verification



Extend verification beyond deck setup

Build from a reliable pre-run check toward broader instrument awareness.

01

Tip counting

Detect missing or depleted tips.

02

Volume monitoring

Use imaging to assess liquid in tips.

03

In-run digital twin

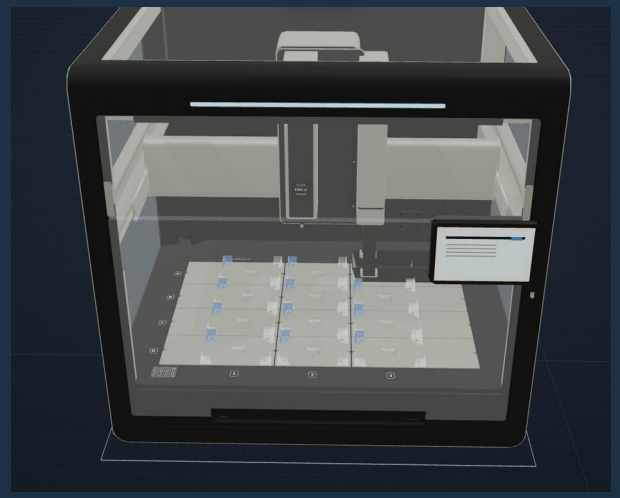
Track tips, liquid and labware during a run.

ROADMAP • Validate the pre-run check, then add monitoring capabilities incrementally.

OpenAI Astra directs the digital twin’s training data

Astra planned every synthetic scene and aimed them at the model’s weak spots; the calibrated twin rendered and labeled them.

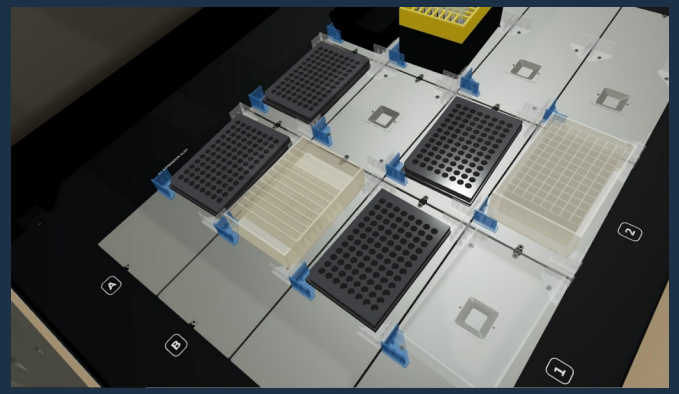
1 · Twin Generated from CAD files and Images



Flex and camera, aligned to the real view

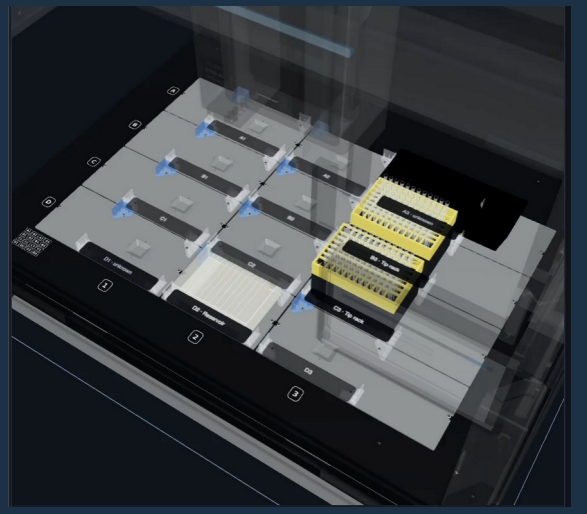
→

2 · Training Images Generated from Twin



→

3 · Live Digital Twin



Detected labware mirrored in 3D

PIPELINE • OpenAI Astra plans 527 scenes → twin renders and auto-labels them → two YOLO26n models → ~0.99 mAP on synthetic validation

THANK YOU · QUESTIONS & DISCUSSION

Lab Sentry

Every run starts on the deck the protocol expects

A live digital twin and a camera check, running locally next to the Opentrons Flex.

Code, setup guide and models: github.com/simoncoelho/lab-sentry

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PHYSICAL DECK ↔ DIGITAL TWIN

